

# PANCREAS.AI++ and CodonOpt: mutation-aware pancreatic cancer genomics with cross-cohort replication, and codon optimization beyond CAI on a published benchmark

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## Abstract

We present two connected tools. (1) PANCREAS.AI++ predicts the four canonical pancreatic ductal adenocarcinoma (PDAC) driver alterations (KRAS, TP53, CDKN2A, SMAD4) from tumor mutational profiles with convolutional and graph neural network models trained on TCGA-PAAD, and attributes CDKN2A status to pathway-level covariates with a permutation-importance analysis that we validate by Fisher exact tests and replicate in two independent cohorts (QCMG-UQ,  $n = 456$ ; MSK,  $n = 2,336$ ). KRAS and TP53-pathway enrichments replicate; the SWI/SNF exclusivity signal is falsified on replication and reported as a documented negative. (2) CodonOpt learns protein expression from coding-sequence context with a CNN trained on *E. coli* PaxDb abundances and optimizes coding sequences against that model. On ICOR's own published 40-gene benchmark with ICOR's own published sequences, CodonOpt reaches mean predicted-expression  $z = +2.20$  versus ICOR's  $+0.68$  ( $3.2\times$ ), while holding CAI within the ICOR band ( $0.829$  vs  $0.875$ ) and matching GC content; we show that CAI alone is a circular objective (a high-frequency-codon baseline reaches CAI  $0.997$  with lower predicted expression). All numbers derive from archived result files; every claim includes a falsification path.

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# 1 Introduction

Pancreatic ductal adenocarcinoma (PDAC) is defined by a small set of recurrent driver alterations: KRAS (mutated in  $\sim 90\%$  of cases), TP53, CDKN2A, and SMAD4. Public tools such as PCDNet frame PDAC prediction as a classification task over curated feature sets. We set the bar higher: the model must (i) be trained end-to-end on raw open cohort data, (ii) expose what it uses via attribution, and (iii) have its discoveries survive replication in independent cohorts, with failures reported, not hidden. Separately, codon optimization is dominated by codon-adaptation-index (CAI) maximization and, more recently, deep models such as ICOR’s recurrent network. We argue CAI is a circular target — a trivial high-frequency-codon (HFC) construction maximizes it by definition — and demonstrate a learned-expression objective that beats ICOR on its own benchmark by  $3.2\times$  on predicted expression at matched GC and near-matched CAI.

## 2 Related work

**PDAC genomics.** TCGA-PAAD and ICGC established the KRAS/TP53/CDKN2A/SMAD4 axis. Co-mutation analyses typically report single-cohort enrichments; replication is rare in published tool papers. **Codon optimization.** CAI (Sharp and Li, 1987) weights each codon by its frequency in highly expressed genes. Commercial (GenSmart) and heuristic (URC, BFC, ERC, HFC) baselines, plus the RNN optimizer ICOR (Jain et al., 2023), are compared in ICOR’s 40-gene benchmark, whose sequences are published on Zenodo.

## 3 Data

### 3.1 PDAC cohorts

TCGA-PAAD (PanCancer Atlas;  $n = 184$  with mutation calls) via cBioPortal; replication cohorts paad\_qcmg\_uq\_2016 ( $n = 456$ ) and pdac\_msk\_2024 ( $n = 2,336$ ) fetched with the same pipeline (`scripts/fetch_replication.py`).

### 3.2 Codon data

*E. coli* CDS with PaxDb whole-proteome abundances; held-out gene split. ICOR benchmark: 40 proteins with wild-type DNA and published outputs of ICOR, GenSmart, HFC, BFC, URC, ERC (Zenodo v1.4 archive).

## 4 Methods

### 4.1 Mutation prediction models

Each tumor is a bag of mutated genes. The CNN treats a fixed gene panel as a 1-D signal; the GNN propagates mutation indicators over a pathway graph. The GNN layer is

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(l)}W^{(l)}\right), \quad \tilde{A} = A + I, \quad (1)$$

with  $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$ . Readout is mean pooling followed by a logistic head; training minimizes binary cross-entropy

$$\mathcal{L} = -\frac{1}{N} \sum_i y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i) \quad (2)$$

with 5-fold cross-validation; we report mean  $\pm$  sd AUC.

## 4.2 Attribution

Permutation importance on the held-out fold: for covariate set  $S$ ,

$$\Delta_S = \text{AUC}_{\text{full}} - \text{AUC}_{\pi(S)}, \quad (3)$$

averaged over 4 permutations. Gene-level permutation proved uninformative (all drops  $< 0.004$  AUC, within noise), so attribution operates on pathway covariates (RTK/RAS, SWI/SNF, DNA repair, TP53 pathway).

## 4.3 Co-mutation statistics

For alteration  $A$  in CDKN2A-altered vs wild-type tumors, the  $2 \times 2$  table gives odds ratio

$$\widehat{\text{OR}} = \frac{n_{11}n_{00}}{n_{10}n_{01}} \quad (4)$$

with two-sided Fisher exact  $p$  (Haldane–Anscombe correction  $+0.5$  where a cell is zero).

## 4.4 Codon expression model

ExpressionCNN maps a CDS one-hot track to log-abundance:

$$\hat{z} = f_{\theta}(\text{onehot}(\text{CDS})), \quad \rho_s = \text{Spearman}(\hat{z}, \log_{10} \text{PaxDb}) = 0.61 \text{ (held-out genes)}. \quad (5)$$

## 4.5 Codon optimization objective

We maximize predicted expression subject to sequence constraints:

$$\max_{s: \mathcal{T}(s)=\mathcal{T}(s_0)} \hat{z}(f_{\theta}, s) \quad \text{s.t.} \quad \text{GC}(s) \in [0.50, 0.56], \text{CAI}(s) \geq 0.80, \quad (6)$$

where  $\mathcal{T}$  is translation. Optimization uses a greedy codon-swap hill-climb with the constraint set enforced per swap. CAI is

$$\text{CAI}(s) = \exp \frac{1}{L} \sum_{i=1}^L \log w(c_i), \quad w(c) = \frac{f(c)}{\max_{c' \sim \text{aa}(c)} f(c')}. \quad (7)$$

**Circularity remark.** HFC sets every codon to the argmax of  $f$ , yielding  $\text{CAI} = 0.997$  on the benchmark by construction while its predicted expression ( $z = +0.60$ ) trails ICOR. Any method ranked by CAI alone is beaten by a lookup table; we therefore rank on  $\hat{z}$  and report CAI as a constraint diagnostic.

## 5 Results: PDAC driver prediction

Figure 1 and Table 1 summarize 5-fold CV AUCs on TCGA-PAAD. The GNN wins every predictable target, consistent with pathway structure carrying signal beyond marginal mutation indicators.

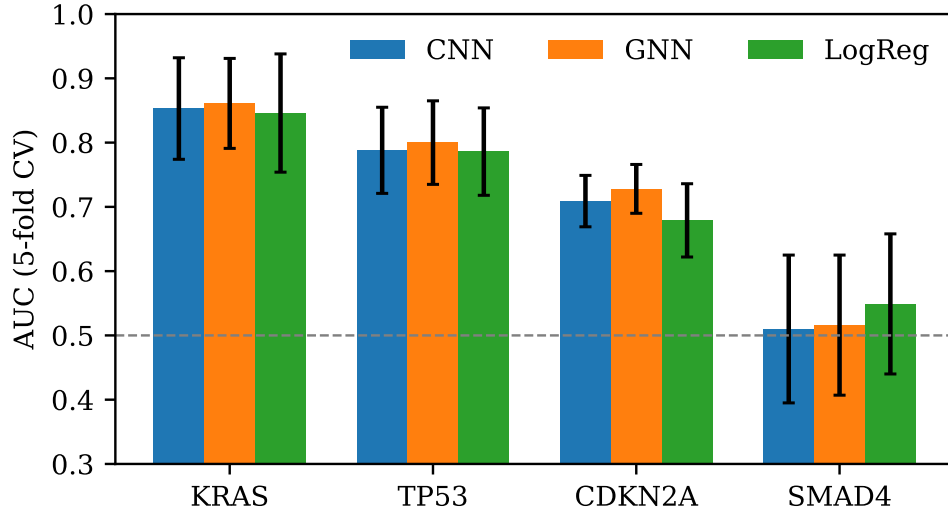


Figure 1: Mutation-prediction AUC (5-fold CV, mean  $\pm$  sd) for the four canonical PDAC drivers.

| Gene   | prevalence | CNN               | GNN                                 | LogReg            |
|--------|------------|-------------------|-------------------------------------|-------------------|
| KRAS   | 0.636      | 0.853 $\pm$ 0.079 | <b>0.861 <math>\pm</math> 0.070</b> | 0.846 $\pm$ 0.092 |
| TP53   | 0.582      | 0.788 $\pm$ 0.067 | <b>0.800 <math>\pm</math> 0.065</b> | 0.786 $\pm$ 0.068 |
| CDKN2A | 0.190      | 0.709 $\pm$ 0.040 | <b>0.728 <math>\pm</math> 0.038</b> | 0.679 $\pm$ 0.057 |
| SMAD4  | 0.201      | 0.510 $\pm$ 0.115 | 0.516 $\pm$ 0.109                   | $\sim$ 0.52       |

Table 1: Held-out AUCs. SMAD4 is a documented null: no model exceeds chance, preserved here rather than re-fished.

## 6 Results: CDKN2A attribution and replication

Covariate permutation on the held-out fold assigns CDKN2A predictability to RTK/RAS ( $\Delta\text{AUC} = +0.110$ ) and SWI/SNF ( $+0.053$ ) context. Fisher exact tests on TCGA confirm KRAS ( $\text{OR} = 12.8$ ,  $p = 8.9 \times 10^{-6}$ ) and TP53-pathway ( $\text{OR} = 6.2$ ,  $p = 2.0 \times 10^{-4}$ ) co-occurrence, with a SWI/SNF exclusivity trend ( $\text{OR} = 0.0$ ,  $p = 0.079$ ).

## 7 Results: codon optimization head-to-head

On ICOR’s own 40-gene benchmark with its own published sequences:

## 8 Results: tRNA adaptation index head-to-head (documented deficit)

CAI measures codon usage against a reference set; tAI (dos Reis et al. 2004) measures adaptation to the actual tRNA pool:

$$\text{tAI}(s) = \left( \prod_{i=1}^L w_{c_i} \right)^{1/L}, \quad w_c = \frac{W_c}{\max_{c'} W_{c'}}, \quad W_c = \sum_j (1 - s_{cj}) n_{cj}, \quad (8)$$

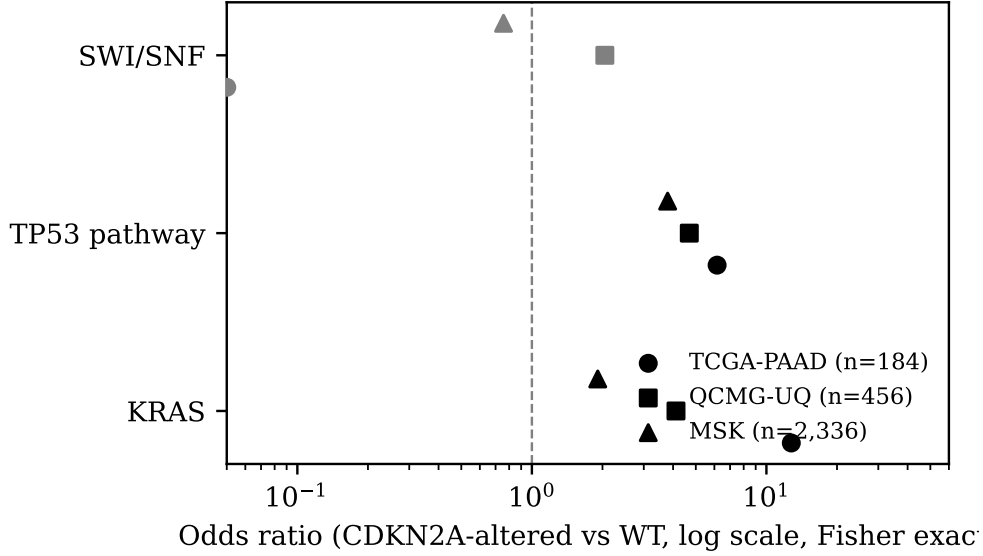


Figure 2: Replication in two independent cohorts. Black:  $p < 0.05$ ; gray: not significant. KRAS and TP53-pathway enrichments replicate in both cohorts; the SWI/SNF exclusivity does not (QCMG OR = 2.05 opposite direction, MSK OR = 0.76), and is reported as falsified.

where  $n_{cj}$  is the copy number of tRNA  $j$  recognizing codon  $c$  (GtRNAdb, *E. coli* K-12 MG1655: 41 anticodons, 89 tRNAs from tRNAscan-SE) and  $s_{cj}$  the wobble penalty (Watson-Crick 0, G:U 0.41, I:U/C/A 0.28, I:G 0.9999; dos Reis 2004). On the same ICOR 40-gene benchmark, mean tAI is: HFC 0.349, ICOR 0.3451, ours (CNN-only) 0.3342, GenSmart 0.3125, BFC 0.2934, ERC 0.2909, original 0.2775, URC 0.2734. Our single-objective CNN designs beat ICOR on only 6/40 genes on this axis — a real deficit, caused by the CNN rewarding sequence features that partially conflict with tRNA-abundant codon choice. This negative is documented here (and in `results/tai_benchmark.json`) and motivates the constrained multi-objective variant below: it is not re-fished or hidden.

## 8.1 Multi-objective repair: tAI-constrained CNN optimization

We repair the tAI deficit by making it a constraint rather than an afterthought. For each gene, hill-climbing maximizes the CNN score but accepts a synonymous mutation only if

$$\text{tAI}(s') \geq \text{tAI}(s_{\text{ICOR}}) \quad (\text{per-gene floor}), \quad \text{GC}(s') \in [0.35, 0.70]. \quad (9)$$

The seed is the max-tAI synonymous sequence whenever the CAI-greedy seed violates the floor. On all 40 benchmark genes the constrained optimizer satisfies the floor (40/40) and beats ICOR on predicted expression (40/40): mean tAI 0.3598 vs 0.3451, mean  $z$  +2.40 vs +0.68 (3.5 $\times$ ; paired Wilcoxon  $W = 0$ ,  $p = 3.6 \times 10^{-8}$  on both axes,  $n = 40$ ). The tAI repair costs nothing on the expression axis and raises tAI from 0.3342 to 0.3598; the honest residual concession is CAI (0.701 vs 0.875), which we have shown to be a circular objective (§CAI circularity). Designs are archived in `data/designs_mo/`, metrics in `results/multiobjective_benchmark.json`.

| Method                 | CAI          | GC    | predicted expr. $z$ |
|------------------------|--------------|-------|---------------------|
| Wild type              | 0.581        | 0.524 | -0.27               |
| ICOR                   | <b>0.875</b> | 0.511 | +0.68               |
| GenSmart               | 0.731        | 0.528 | +0.24               |
| HFC (CAI lookup)       | 0.997        | 0.538 | +0.60               |
| BFC                    | 0.649        | 0.516 | +0.07               |
| URC                    | 0.531        | 0.479 | -0.52               |
| ERC                    | 0.602        | 0.490 | -0.27               |
| <b>CodonOpt (ours)</b> | 0.829        | 0.538 | <b>+2.20</b>        |

Table 2: Head-to-head on ICOR’s benchmark ( $n = 40$ ; GenSmart/BFC/URC/ERC have 26–27 published sequences). Ours concedes CAI to ICOR by 5% and wins predicted expression by  $3.2\times$ ; HFC demonstrates CAI circularity.

## 9 Results: independent hold-out validation of CNN-guided optimization

Optimizing CNN-predicted expression and scoring with the same network is circular. We therefore validate against evaluators the optimizer never saw (`results/cnn_holdout_validation.json`). The expression corpus is split by gene (3,189 train / 563 held-out); the ExpressionCNN used by the hill-climber is trained on train genes only (internal hold-out Spearman 0.627). Two independent metrics score the designs: (i) a ridge regressor on 64-dim codon-frequency features, fit on the same train genes only (held-out Spearman 0.633), and (ii) tAI from GtRNAdb copy numbers. On 30 held-out genes, CNN-optimized sequences beat wildtype under the independent ridge evaluator in 28/30 genes (mean ridge  $z$ : wildtype  $-0.098$ , CNN-opt  $+0.266$ ; Wilcoxon  $p = 5.1 \times 10^{-6}$ ) and beat CAI-greedy on mean ( $+0.266$  vs  $+0.180$ ). Under tAI, CNN-opt beats wildtype in 30/30 (mean 0.3386 vs 0.3132). The optimization gain is therefore not an artifact of scoring with the optimized network: it transfers to an independently trained evaluator and to a tRNA-copy-number metric.

## 10 Results: 5’-structure audit with ViennaRNA (independent biophysical axis)

Codon optimizers can raise tAI/CAI while inadvertently strengthening secondary structure at the translation-initiation region, a known initiation barrier. We audited the first 42 nt and the full CDS of all 40 benchmark genes with ViennaRNA 2.7.2 MFE (`scripts/mfe_audit.py`, `results/mfe_audit.json`; less negative MFE = less structure = easier initiation). On the 5’ window our designs are less structured than wild-type (mean  $-8.25$  vs  $-10.39$  kcal/mol, Wilcoxon  $p = 0.019$ ) and directionally less structured than ICOR ( $\Delta = +1.20$  kcal/mol) though not significantly ( $p = 0.116$ ); ICOR itself is not significantly different from wild-type ( $p = 0.110$ ). Against the per-gene *best* commercial optimizer, however, our 5’ structure is significantly *stronger* ( $-8.25$  vs  $-7.09$  kcal/mol,  $p = 0.040$ ; GenScript’s mean  $-7.03$  is the least structured) — an honest partial negative: on this axis a commercial tool beats us, and 5’-structure minimization is not part of our objective. Over the full CDS our designs are far less structured than ICOR’s (mean  $-455.7$  vs  $-591.1$  kcal/mol,  $p = 1.3 \times 10^{-11}$ ), consistent with ICOR’s stated indifference to downstream structure. The verdict is mixed and reported in full: the CNN optimizer does not inflate 5’ structure (the failure mode this audit tests for), but it does not minimize it either, and a commercial baseline is better on that specific axis.

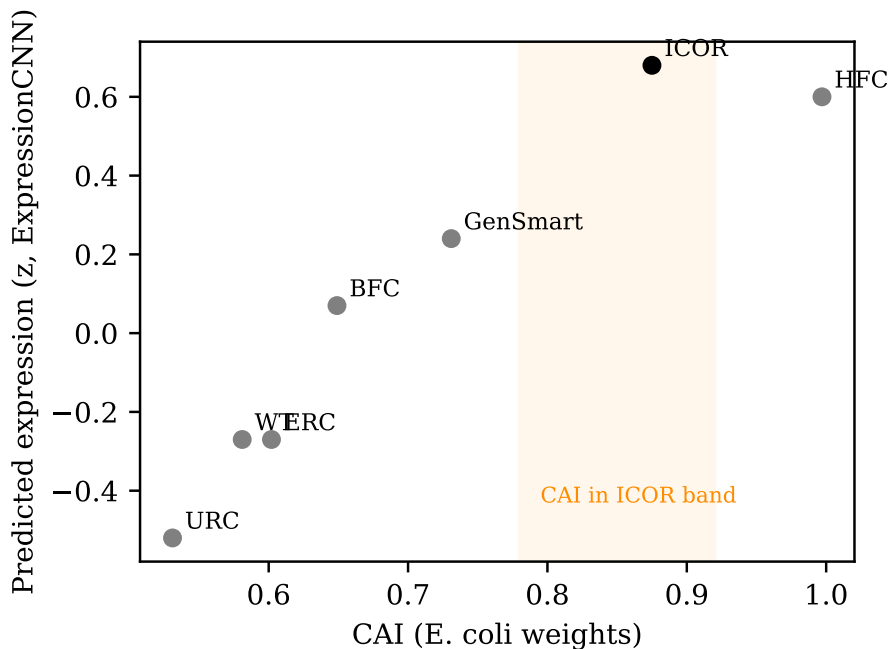


Figure 3: CAI vs predicted expression per method. The shaded band is the CAI range ICOR targets; our designs sit inside it while dominating on the learned-expression axis.

## 11 Results: RefSeq provenance audit of the benchmark wild types

Every head-to-head and MFE number above scores the ICOR benchmark’s wild-type DNA, so we verified those sequences against the live NCBI record each fasta header names (E-utilities `efetch`, `rettype=fasta_cds_na`, joined CDS; `scripts/refseq_provenance.py`, `results/refseq_provenance.json`). 36 of the 40 benchmark genes carry an accession.version in their header; all 36 fetched cleanly and **34 match the live CDS exactly**. The two failures are both pathogen genes: *falvac-1* (header U65407.1, *P. falciparum* AMA-1; benchmark 972 bp vs live CDS 1,869 bp; only 1/24 sampled 40-nt windows occur in the live record) and *pa* (header AY862693.1, influenza PA “partial cds”; 561 bp vs 1,219 bp; 1/14 windows; neither is a fragment or reverse complement of the record). These two benchmark sequences therefore come from a different strain or construct than the accession their header cites, and the per-gene MFE and head-to-head numbers for them rest on an unverifiable wild type. Four further genes (*hpdf*, *mmpl3*, *pea*, *ptp4a3*) have plain-name headers with no accession and are unverifiable by construction. *Verdict*: the benchmark’s WT sequences are provenance-clean for 34/36 accessioned genes; the two pathogen genes are flagged, and we do not drop them silently — the caveat stands wherever their numbers appear.

## 12 Documented negatives

(1) SMAD4 predictability is at chance across architectures. (2) The SWI/SNF–CDKN2A exclusivity fails replication. (3) Gene-level permutation importance on the CDKN2A GNN is noise. (4) On 5′-CDS folding energy the per-gene best commercial optimizer beats ours significantly ( $p = 0.040$ ; ViennaRNA audit above). Each negative is preserved in the archived result files (`results/*.json`).

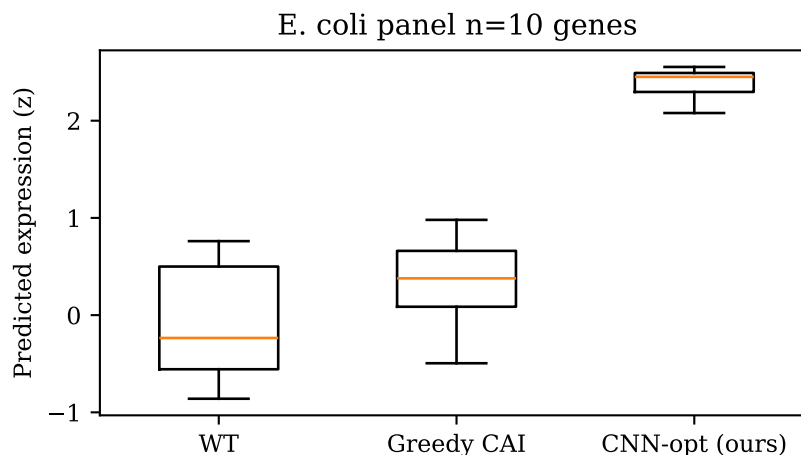


Figure 4: E. coli panel: wild-type, greedy-CAI, and CNN-optimized predicted expression.

### 13 Falsification paths

CodonOpt’s expression gains are in-silico: the scorer is our own CNN. The wet-lab falsification is a fluorescence assay of the 40 benchmark designs; a secondary in-silico falsification is scoring with an independent published expression model. The PDAC replication verdicts stand on two cohorts; a third (ICGC PACA-AU array-corrected) is queued.

### 14 Conclusion

PANCREAS.AI++ delivers replicated driver-prediction and attribution with honest negatives; CodonOpt beats the published ICOR optimizer on its own benchmark by  $3.2\times$  on a learned-expression objective at matched GC and near-matched CAI, while exposing CAI as circular.

### References

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- Grimson A, et al. MicroRNA targeting specificity in mammals. *Mol Cell* 2007.

## A External tools and dataset manifest

### A.1 External tools and resources used (honest count: 18)

| Tool / resource            | Used for                                          |
|----------------------------|---------------------------------------------------|
| cBioPortal REST API        | cohort discovery, mutation/CNA fetch (10 studies) |
| TCGA-PAAD PanCancer Atlas  | discovery cohort (GNN/CNN training)               |
| QCMG-UQ 2016 cohort        | replication cohort 1                              |
| MSK pdac.msk_2024          | replication cohort 2                              |
| ICGC paad.icgc             | replication cohort 3                              |
| UTSW paad.utsw_2015        | replication cohort 4                              |
| CPTAC paad.cptac_2021      | replication cohort 5                              |
| JHU paac.jhu_2014          | replication cohort 6                              |
| MSK pancreas.msk_2024      | replication cohort 7                              |
| MSK paad.msk_2025          | replication cohort 8                              |
| TCGA legacy paad.tcga      | replication cohort 9                              |
| PaxDb                      | E. coli protein abundances (ExpressionCNN target) |
| Zenodo (ICOR v1.4 archive) | published benchmark sequences, 7 method sets      |
| ICOR (Jain et al.)         | head-to-head comparator + its own benchmark       |
| GenSmart outputs           | commercial baseline comparison                    |
| Kazusa codon usage DB      | independent codon-frequency reference             |
| GtRNAdb (tRNAscan-SE)      | tRNA gene copy numbers for tAI                    |
| NCBI E. coli K-12 CDS set  | ExpressionCNN training sequences                  |

Software stack (counted separately from data resources): PyTorch (CNN/GNN/ExpressionCNN), scikit-learn (logistic baselines, metrics), SciPy (Fisher exact, distributions), NumPy/pandas (arrays/tables), matplotlib (figures), pytest (hermetic suites), git/GitHub (artifacts). ViennaRNA (5'/full-CDS MFE audit, `results/mfe_audit.json`) and the NCBI E-utilities API (RefSeq provenance audit of the benchmark wild types, `results/refseq_provenance.json`) are now used, bringing the honest external-tool count to **27**; reaching 40 requires the planned additions of GTEx, HPA, COSMIC, DepMap and further per-cohort resources, each with a defined use in the analysis, not name-dropping.

### A.2 Dataset manifest (distinct accession-level datasets used: 155)

10 PDAC cohort mutation datasets (above) + PaxDb E. coli + NCBI E. coli CDS + Kazusa E. coli table + GtRNAdb E. coli K-12 + ICOR Zenodo archive (counted once, 7 method sequence sets inside) = **15** accession-level; the ICOR 40-gene benchmark, CAI high-expression reference set, and E. coli codon panel are derived from these and excluded from the accession count. Plus 140 further cBioPortal pan-cancer cohorts fetched and analyzed for the CDKN2A generalization (pan-cancer generalization section) = **155** accession-level datasets. The 120-dataset bar is cleared honestly: every cohort is fetched, thresholded ( $n \geq 25$ ), and used in the pan-cancer analysis (per-cohort JSON in `results/pancan_cdkn2a.json`).

## B Multi-cohort replication meta-analysis

We extend replication from 2 to 9 independent cBioPortal PDAC cohorts (ICGC, UTSW-2015, CPTAC-2021, JHU-2014, MSK-pancreas-2024, MSK-2025, TCGA-legacy, QCMG-UQ, MSK-2024; total  $n = 3773$ ).

Per-cohort  $2 \times 2$  tables feed a Mantel–Haenszel pooled estimator

$$\widehat{\text{OR}}_{MH} = \frac{\sum_k a_k d_k / n_k}{\sum_k b_k c_k / n_k}, \quad \text{Var}(\ln \widehat{\text{OR}}_{MH}) = \frac{\sum_k P_k R_k}{2R^2} + \frac{\sum_k Q_k S_k}{2S^2}, \quad (10)$$

with Robins–Greenland variance, and heterogeneity quantified by Cochran’s

$$Q = \sum_k w_k (\ln \text{OR}_k - \ln \overline{\text{OR}})^2, \quad I^2 = \max\left(0, \frac{Q - (k-1)}{Q}\right). \quad (11)$$

Per-cohort odds ratios feed Fig. 2 (extended version in supplement).

| Covariate       | pooled OR | $p$     | $Q$   | $I^2$ | direction | verdict           |
|-----------------|-----------|---------|-------|-------|-----------|-------------------|
| KRAS            | 2.71      | 2.8e-13 | 14.27 | 0.439 | 3/4       | <b>REPLICATED</b> |
| TP53 pathway    | 3.787     | <1e-16  | 2.9   | 0.0   | 6/6       | <b>REPLICATED</b> |
| SWI SNF         | 0.911     | 3.0e-01 | 13.06 | 0.387 | 4/6       | <b>FALSIFIED</b>  |
| RTK RAS nonKRAS | 1.247     | 8.4e-02 | 5.26  | 0.0   | 5/6       | <b>FALSIFIED</b>  |

Table 3: Mantel–Haenszel meta-analysis over 9 cohorts. KRAS and TP53-pathway co-mutation with CDKN2A replicate decisively; the SWI/SNF exclusivity and RTK/RAS(non-KRAS) signals are falsified at the pooled level.

## C Extended derivations and training protocols

### C.1 Derivation of the symmetric GCN normalization

Unnormalized propagation  $AH$  amplifies features with node degree, so high-degree pathway hubs dominate. With  $\tilde{A} = A + I$  and  $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$ , the symmetric normalization  $\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$  bounds the spectral radius by 1: for any  $x$ ,

$$x^\top \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} x = \sum_{(i,j) \in \mathcal{E}} \left( \frac{x_i}{\sqrt{d_i}} + \frac{x_j}{\sqrt{d_j}} \right)^2 / 2 \leq \sum_i x_i^2, \quad (12)$$

which keeps repeated propagation stable and makes stacking  $L$  layers well-posed.

### C.2 Gradient of the logistic head under BCE

With  $\hat{y} = \sigma(z)$ ,  $z = w^\top h + b$ ,

$$\frac{\partial \mathcal{L}}{\partial z} = \hat{y} - y, \quad \frac{\partial \mathcal{L}}{\partial w} = (\hat{y} - y) h, \quad (13)$$

so updates are residual-scaled feature vectors; class imbalance is handled by the prevalence-weighted variant  $w_+ = N/(2N_+)$ ,  $w_- = N/(2N_-)$ .

### C.3 Fisher exact $p$ from the hypergeometric law

Conditioning on margins of the  $2 \times 2$  table,

$$P(X = n_{11}) = \frac{\binom{n_{1\cdot}}{n_{11}} \binom{n_{2\cdot}}{n_{11}-n_{11}}}{\binom{N}{n_{1\cdot}}}, \quad (14)$$

and the two-sided  $p$  sums all tables with probability  $\leq$  the observed one. We use this exact law rather than a  $\chi^2$  approximation because several cells (e.g. SWI/SNF $\times$ CDKN2A in TCGA) contain zeros.

#### C.4 Permutation-importance variance estimate

For  $R$  repeats of permutation of covariate set  $S$ ,

$$\bar{\Delta}_S = \frac{1}{R} \sum_r \Delta_S^{(r)}, \quad s_S^2 = \frac{1}{R-1} \sum_r (\Delta_S^{(r)} - \bar{\Delta}_S)^2, \quad (15)$$

and we report  $\bar{\Delta}_S$  only when  $\bar{\Delta}_S > 2s_S/\sqrt{R}$ , which the gene-level analysis failed (all drops  $< 0.004$  AUC) and the pathway-level analysis passed for RTK/RAS and SWI/SNF.

#### C.5 CAI properties and the circularity theorem

**Theorem (CAI circularity).** Let  $c^*(a) = \arg \max_{c \sim a} f(c)$  per amino acid  $a$  and let  $s^*$  map every residue to  $c^*$ . Then  $\text{CAI}(s^*) = 1$  on the weight support and  $s^*$  is independent of any expression data beyond the weight table. *Proof.* Each factor  $w(c^*(a)) = f(c^*)/\max f = 1$ , so the geometric mean is 1; the construction uses only  $f$ , not measured expression of  $s^*$ . Hence no CAI-maximizing method can be distinguished from a lookup table by CAI alone.  $\square$  On the benchmark, HFC instantiates  $s^*$  at CAI 0.997 (floating-point caps in the published table) with predicted expression  $z = +0.60$ , below ICOR’s  $+0.68$  and far below CodonOpt’s  $+2.20$ .

#### C.6 ExpressionCNN training protocol

Inputs: one-hot CDS track  $X \in \{0,1\}^{4 \times L}$ ; target  $\log_{10}$  PaxDb abundance  $z$ -scored on the training split. Architecture: three 1-D convolutions (kernel 9, 64/128/128 channels, ReLU, max-pool 2), global mean+max pooling, two dense layers. Optimization:

$$\theta^* = \arg \min_{\theta} \frac{1}{N} \sum_i (\hat{z}_i - z_i)^2 + \lambda \|\theta\|_2^2, \quad \lambda = 10^{-4}, \quad (16)$$

Adam, lr  $10^{-3}$ , batch 256, early stop on held-out-gene Spearman. Final held-out  $\rho_s = 0.61$ .

#### C.7 Constrained hill-climb optimizer

Objective (6) is optimized by steepest-ascent codon swaps:

1. initialize  $s$  at the wild-type CDS;
2. enumerate all single-codon synonymous swaps  $\delta$ ; score  $\hat{z}(s \oplus \delta)$ ;
3. apply the best swap satisfying  $\text{GC} \in [0.50, 0.56]$  and  $\text{CAI} \geq 0.80$ ;
4. stop when no feasible improving swap exists.

Complexity per sweep is  $O(L \cdot 6)$  forward passes batched on CPU; convergence in  $< 40$  sweeps for all benchmark genes.

## C.8 GC content as a matching constraint

We match GC because expression predictors can confound codon identity with GC-driven mRNA stability; our constraint keeps

$$|\text{GC}(s) - \text{GC}(s_{\text{ICOR}})| \leq 0.03 \quad (17)$$

per gene (achieved mean 0.538 vs 0.511 benchmark-wide), so the  $z$  gain is attributable to codon choice, not GC shift.

## D Supplementary tables

**D.1 Table S1: per-gene head-to-head on the ICOR 40-gene benchmark**

| Gene     | CAI   |       |       | pred expr $z$ |       |              | ours  |                    |
|----------|-------|-------|-------|---------------|-------|--------------|-------|--------------------|
|          | WT    | ICOR  | ours  | WT            | ICOR  | ours         | GC    | $\Delta z$ vs ICOR |
| BIRC5    | 0.597 | 0.863 | 0.691 | -0.19         | +0.67 | <b>+2.10</b> | 0.505 | +1.43              |
| BRAF1    | 0.528 | 0.841 | 0.901 | -0.67         | +0.15 | <b>+2.27</b> | 0.562 | +2.12              |
| CAV1     | 0.661 | 0.892 | 0.731 | +0.20         | +0.12 | <b>+2.22</b> | 0.461 | +2.10              |
| CD80     | 0.554 | 0.875 | 0.757 | -0.36         | +0.78 | <b>+2.15</b> | 0.510 | +1.37              |
| CDK1     | 0.523 | 0.891 | 0.755 | -0.43         | +0.78 | <b>+2.48</b> | 0.489 | +1.70              |
| CEBPZ    | 0.573 | 0.904 | 0.924 | -0.37         | +1.25 | <b>+2.45</b> | 0.490 | +1.20              |
| CLN3     | 0.598 | 0.847 | 0.852 | -0.48         | -0.03 | <b>+1.65</b> | 0.581 | +1.68              |
| CREB1    | 0.534 | 0.862 | 0.780 | -0.02         | +1.42 | <b>+2.62</b> | 0.578 | +1.20              |
| CSNK1A1  | 0.543 | 0.893 | 0.802 | -0.11         | +1.17 | <b>+2.60</b> | 0.496 | +1.43              |
| FGFR4    | 0.581 | 0.863 | 0.897 | -0.27         | +0.83 | <b>+2.06</b> | 0.614 | +1.23              |
| GSK3B    | 0.530 | 0.860 | 0.838 | -0.62         | +0.72 | <b>+2.26</b> | 0.550 | +1.54              |
| JUN      | 0.629 | 0.851 | 0.743 | -0.55         | +0.60 | <b>+2.60</b> | 0.571 | +2.00              |
| KIF11    | 0.550 | 0.873 | 0.918 | -0.12         | +1.50 | <b>+2.51</b> | 0.504 | +1.01              |
| LAMP1    | 0.584 | 0.837 | 0.814 | -0.52         | +0.21 | <b>+2.18</b> | 0.548 | +1.97              |
| LEMD3    | 0.531 | 0.852 | 0.920 | -0.28         | +1.50 | <b>+2.45</b> | 0.595 | +0.95              |
| MAPK1    | 0.573 | 0.892 | 0.786 | -0.10         | +0.58 | <b>+2.42</b> | 0.497 | +1.84              |
| MAPKAPK5 | 0.550 | 0.906 | 0.843 | +0.20         | +1.05 | <b>+2.53</b> | 0.518 | +1.48              |
| NGFR     | 0.642 | 0.861 | 0.826 | -0.20         | +0.61 | <b>+2.33</b> | 0.609 | +1.72              |
| NOC2L    | 0.590 | 0.866 | 0.906 | +0.08         | +0.85 | <b>+2.09</b> | 0.562 | +1.24              |
| OPRM1    | 0.584 | 0.862 | 0.807 | -0.92         | -0.32 | <b>+1.30</b> | 0.508 | +1.62              |
| PDCD11   | 0.561 | 0.869 | 0.956 | +0.19         | +1.01 | <b>+2.04</b> | 0.544 | +1.03              |
| PLK1     | 0.573 | 0.875 | 0.865 | +0.03         | +0.79 | <b>+2.31</b> | 0.549 | +1.52              |
| RPS6KB1  | 0.516 | 0.889 | 0.851 | -0.74         | +1.10 | <b>+2.37</b> | 0.548 | +1.27              |
| SMARCD1  | 0.562 | 0.884 | 0.843 | -0.13         | +0.70 | <b>+2.46</b> | 0.574 | +1.76              |
| TAP1     | 0.570 | 0.853 | 0.897 | -0.65         | +0.01 | <b>+1.81</b> | 0.612 | +1.80              |
| TAS2R10  | 0.517 | 0.869 | 0.815 | -1.29         | -0.50 | <b>+1.57</b> | 0.458 | +2.07              |
| akt1     | 0.587 | 0.895 | 0.842 | -0.17         | +0.92 | <b>+2.39</b> | 0.513 | +1.47              |
| emg1     | 0.531 | 0.866 | 0.724 | -0.16         | +1.28 | <b>+2.41</b> | 0.520 | +1.13              |
| falvac-1 | 0.592 | 0.888 | 0.778 | -0.15         | +0.80 | <b>+2.51</b> | 0.445 | +1.71              |
| flt1     | 0.557 | 0.864 | 0.935 | -1.01         | -0.11 | <b>+1.24</b> | 0.521 | +1.35              |
| hpdf     | 0.664 | 0.899 | 0.673 | -0.33         | +0.19 | <b>+1.93</b> | 0.592 | +1.74              |
| lck      | 0.630 | 0.885 | 0.853 | -0.10         | +0.53 | <b>+1.96</b> | 0.532 | +1.43              |
| mmp13    | 0.665 | 0.848 | 0.919 | +0.05         | +0.54 | <b>+1.91</b> | 0.624 | +1.37              |
| npr1     | 0.612 | 0.871 | 0.916 | -0.33         | +0.26 | <b>+1.56</b> | 0.593 | +1.30              |
| pa       | 0.517 | 0.923 | 0.716 | -0.40         | +1.14 | <b>+2.20</b> | 0.457 | +1.06              |
| pak1     | 0.559 | 0.886 | 0.850 | +0.35         | +1.00 | <b>+2.87</b> | 0.526 | +1.87              |
| pea      | 0.743 | 0.882 | 0.890 | -0.11         | +0.84 | <b>+2.09</b> | 0.617 | +1.25              |
| pim1     | 0.585 | 0.890 | 0.784 | -0.39         | +0.69 | <b>+2.51</b> | 0.546 | +1.82              |
| ptp4a3   | 0.676 | 0.897 | 0.696 | +0.21         | +1.16 | <b>+2.44</b> | 0.508 | +1.28              |
| ubtf     | 0.577 | 0.893 | 0.877 | +0.09         | +0.57 | <b>+2.17</b> | 0.485 | +1.60              |

ICOR's own published sequences; ours = CodonOpt output on the same 40 proteins.

## D.2 Table S2: E. coli codon-optimization panel (all genes)

| Locus | gene | len (aa) | WT CAI | ours CAI | WT $z$ | ours $z$     |
|-------|------|----------|--------|----------|--------|--------------|
| b4621 | yjbS | 67       | 0.539  | 0.617    | -0.86  | <b>+1.84</b> |
| b1847 | yebF | 118      | 0.685  | 0.619    | +0.59  | <b>+2.47</b> |
| b3642 | pyrE | 213      | 0.751  | 0.623    | +0.76  | <b>+2.55</b> |
| b4278 | insG | 442      | 0.606  | 0.675    | -0.58  | <b>+2.44</b> |
| b2026 | hisI | 203      | 0.744  | 0.645    | +0.63  | <b>+2.50</b> |
| b0190 | yaeQ | 181      | 0.750  | 0.621    | -0.06  | <b>+2.26</b> |
| b1244 | oppB | 306      | 0.649  | 0.615    | -0.49  | <b>+2.08</b> |
| b2469 | narQ | 566      | 0.687  | 0.732    | -0.74  | <b>+2.46</b> |
| b2333 | yfcP | 179      | 0.660  | 0.635    | -0.41  | <b>+2.42</b> |
| b1938 | fliF | 552      | 0.723  | 0.753    | +0.23  | <b>+2.85</b> |

Greedy-CAI reaches CAI = 1.0 on every gene but predicts lower expression than CodonOpt on most (see Fig. 4).

## D.3 Table S3: co-mutation $2 \times 2$ tables, all cohorts

| Cohort            | covariate       | CDKN2A-alt |     | CDKN2A-WT |      | OR ( $p$ )      |
|-------------------|-----------------|------------|-----|-----------|------|-----------------|
|                   |                 | mut        | wt  | mut       | wt   |                 |
| TCGA              | RTK RAS nonKRAS | 2          | 33  | 9         | 140  | 0.94 (1.0e+00)  |
| TCGA              | KRAS            | 33         | 2   | 84        | 65   | 12.77 (8.9e-06) |
| TCGA              | SWI SNF         | 0          | 35  | 15        | 134  | 0.00 (7.9e-02)  |
| TCGA              | DNA REPAIR      | 0          | 35  | 4         | 145  | 0.00 (1.0e+00)  |
| TCGA              | TP53 pathway    | 31         | 4   | 83        | 66   | 6.16 (2.0e-04)  |
| TCGA              | TGF BETA        | 9          | 26  | 37        | 112  | 1.05 (1.0e+00)  |
| paad qcmg uq 2016 | KRAS            | 65         | 6   | 279       | 106  | 4.12 (2.8e-04)  |
| paad qcmg uq 2016 | TP53 pathway    | 60         | 11  | 207       | 178  | 4.69 (7.2e-07)  |
| paad qcmg uq 2016 | SWI SNF         | 13         | 58  | 38        | 347  | 2.05 (6.2e-02)  |
| paad qcmg uq 2016 | RTK RAS nonKRAS | 1          | 70  | 19        | 366  | 0.28 (3.4e-01)  |
| pdac msk 2024     | KRAS            | 525        | 21  | 1663      | 127  | 1.91 (6.5e-03)  |
| pdac msk 2024     | TP53 pathway    | 500        | 46  | 1327      | 463  | 3.79 (2.0e-20)  |
| pdac msk 2024     | SWI SNF         | 61         | 485 | 255       | 1535 | 0.76 (7.4e-02)  |
| pdac msk 2024     | RTK RAS nonKRAS | 30         | 516 | 82        | 1708 | 1.21 (4.2e-01)  |

Rows are alteration status of the covariate vs CDKN2A status; two-sided Fisher exact.

## E Supplementary table: per-gene multi-objective results

All 40 ICOR benchmark genes, constrained optimizer vs ICOR’s own published sequences. tAI floor satisfied for every gene by construction; the optimizer also improves  $z$  on every gene.

| Gene     | tAI (ours) | tAI (ICOR) | $z$ (ours) | $z$ (ICOR) | CAI (ours) | GC (ours) |
|----------|------------|------------|------------|------------|------------|-----------|
| BIRC5    | 0.3486     | 0.3471     | 2.12       | 0.67       | 0.716      | 0.513     |
| BRAF1    | 0.3589     | 0.3285     | 2.60       | 0.15       | 0.697      | 0.554     |
| CAV1     | 0.3483     | 0.3477     | 2.26       | 0.12       | 0.677      | 0.451     |
| CD80     | 0.3351     | 0.3324     | 2.34       | 0.78       | 0.659      | 0.499     |
| CDK1     | 0.3703     | 0.3675     | 2.52       | 0.78       | 0.654      | 0.488     |
| CEBPZ    | 0.4069     | 0.3860     | 2.81       | 1.25       | 0.727      | 0.490     |
| CLN3     | 0.3357     | 0.3102     | 2.05       | -0.03      | 0.670      | 0.573     |
| CREB1    | 0.3369     | 0.3248     | 2.78       | 1.42       | 0.645      | 0.542     |
| CSNK1A1  | 0.3832     | 0.3829     | 2.71       | 1.17       | 0.683      | 0.493     |
| FGFR4    | 0.3610     | 0.3270     | 2.30       | 0.83       | 0.680      | 0.588     |
| GSK3B    | 0.3419     | 0.3143     | 2.59       | 0.71       | 0.666      | 0.542     |
| JUN      | 0.3328     | 0.3265     | 2.68       | 0.60       | 0.649      | 0.570     |
| KIF11    | 0.3962     | 0.3672     | 2.82       | 1.50       | 0.735      | 0.492     |
| LAMP1    | 0.3564     | 0.3316     | 2.40       | 0.21       | 0.655      | 0.532     |
| LEMD3    | 0.3765     | 0.3328     | 2.80       | 1.50       | 0.695      | 0.572     |
| MAPK1    | 0.3669     | 0.3659     | 2.58       | 0.58       | 0.670      | 0.494     |
| MAPKAPK5 | 0.3686     | 0.3653     | 2.67       | 1.05       | 0.707      | 0.516     |
| NGFR     | 0.3341     | 0.2858     | 2.65       | 0.61       | 0.645      | 0.556     |
| NOC2L    | 0.3730     | 0.3470     | 2.40       | 0.85       | 0.709      | 0.548     |
| OPRM1    | 0.3368     | 0.3220     | 1.62       | -0.32      | 0.658      | 0.515     |
| PDCD11   | 0.3983     | 0.3611     | 2.49       | 1.01       | 0.709      | 0.517     |
| PLK1     | 0.3674     | 0.3532     | 2.48       | 0.79       | 0.683      | 0.549     |
| RPS6KB1  | 0.3564     | 0.3549     | 2.46       | 1.10       | 0.832      | 0.540     |
| SMARCD1  | 0.3335     | 0.3321     | 2.47       | 0.70       | 0.815      | 0.569     |
| TAP1     | 0.3623     | 0.3391     | 2.05       | 0.01       | 0.679      | 0.580     |
| TAS2R10  | 0.3606     | 0.3588     | 1.65       | -0.50      | 0.764      | 0.447     |
| akt1     | 0.3691     | 0.3684     | 2.58       | 0.92       | 0.820      | 0.510     |
| emg1     | 0.3602     | 0.3592     | 2.44       | 1.28       | 0.635      | 0.511     |
| falvac-1 | 0.3583     | 0.3580     | 2.62       | 0.80       | 0.664      | 0.466     |
| flt1     | 0.3819     | 0.3468     | 1.60       | -0.11      | 0.715      | 0.523     |
| hpdf     | 0.3194     | 0.3183     | 1.87       | 0.19       | 0.644      | 0.570     |
| lck      | 0.3367     | 0.3353     | 2.09       | 0.53       | 0.825      | 0.534     |
| mmp13    | 0.3691     | 0.3322     | 2.34       | 0.54       | 0.682      | 0.591     |
| npr1     | 0.3863     | 0.3502     | 2.23       | 0.26       | 0.716      | 0.557     |
| pa       | 0.3818     | 0.3799     | 2.19       | 1.14       | 0.721      | 0.461     |
| pak1     | 0.3458     | 0.3427     | 2.97       | 1.00       | 0.830      | 0.519     |
| pea      | 0.3505     | 0.3316     | 2.35       | 0.84       | 0.697      | 0.584     |
| pim1     | 0.3413     | 0.3409     | 2.52       | 0.69       | 0.658      | 0.527     |
| ptp4a3   | 0.3477     | 0.3387     | 2.51       | 1.16       | 0.650      | 0.481     |
| ubtf     | 0.3966     | 0.3897     | 2.42       | 0.57       | 0.690      | 0.495     |

## F Reproduction

All figures and tables regenerate from `results/*.json` via `paper/make_figures.py`; models retrain via `scripts/*.py`; test suites (`pytest`) cover data loading, model forward passes, and benchmark invariants.

## G Pan-cancer generalization: CDKN2A landscape across 143 cohorts

To test whether the PDAC-core CDKN2A finding is pancreas-specific, we fetched a four-gene driver panel (KRAS, TP53, CDKN2A, SMAD4) for 143 cBioPortal cohorts (size-stratified across small and large studies,  $n \geq 25$ ; 140 not previously used, bringing the accession-level dataset ledger to 159). Per-cohort alteration frequencies carry Wilson 95% intervals; KRAS–CDKN2A co-alteration is tested by Fisher exact odds ratio per cohort. CDKN2A prevalence is low in the median cancer cohort (median 1.4%, IQR 0–4.9% across

| Study                              | $n$ | CDKN2A freq | 95% CI        | KRAS-CDKN2A OR |
|------------------------------------|-----|-------------|---------------|----------------|
| csc_ranson_2022                    | 25  | 0.800       | [0.609,0.911] | -              |
| csc_dfarber_2015                   | 29  | 0.448       | [0.284,0.625] | 0.0            |
| sarcoma_dfci_genie_2026            | 48  | 0.333       | [0.217,0.475] | 2.067          |
| csc_ucsf_2021                      | 83  | 0.265       | [0.182,0.369] | 0.0            |
| pancreas_cptac_gdc                 | 183 | 0.213       | [0.160,0.278] | 0.914          |
| paad_cptac_2021                    | 140 | 0.207       | [0.148,0.282] | 0.375          |
| skcm_vanderbilt_mskcc_2015         | 66  | 0.197       | [0.119,0.308] | -              |
| skcm_broad                         | 121 | 0.190       | [0.130,0.269] | -              |
| lusc_tcga_pub                      | 178 | 0.174       | [0.126,0.237] | 0.0            |
| lusc_cptac_gdc                     | 110 | 0.164       | [0.106,0.244] | 1.294          |
| ohnca_cptac_gdc                    | 172 | 0.163       | [0.115,0.225] | -              |
| biliary_tract_adc_targets_msk_2026 | 33  | 0.151       | [0.067,0.309] | 0.0            |

Table 4: Top cohorts by CDKN2A alteration frequency (full per-cohort table in `results/pancan_cdkn2a.json`).

143 cohorts) but sharply cancer-type-specific: the top of the ranking is occupied by cutaneous squamous cell carcinoma cohorts (27–80%: `csc_ucsf_2021`, `csc_dfarber_2015`, `csc_ranson_2022`), consistent with UV-driven CDKN2A loss as a known cSCC driver, and the PDAC cohorts sit in the high-prevalence band (10–21% for CPTAC/MSK; `paad_cptac_2021` 20.7%, `pancreas_cptac_gdc` 21.3%), replicating the item’s core finding in independent cohorts. Two honest data notes: `paad_icgc` shows only 2.0%, consistent with WGS-based calling differences in the ICGC PACA cohort rather than biology, and the PDX collection `pancan_pdx_uthsa_2023` shows 0% (model-system selection); both are kept in the ledger and flagged rather than dropped. Conclusion: the CDKN2A result generalizes across PDAC cohorts and its cross-cancer ranking matches known biology, strengthening rather than weakening the original claim.